Low code Exploratory Data Tools

Created: 03/15/2023
Updated: 03/17/2023

(**Must be executed again in Google Colab, to show all outputs**).

This Google Colab Jupyter Notebook shows EDA examples using a set of diverse low-code exploratory data tools:

- ydata-profiling, also known as `pandas-profiling`, the primary goal is to provide a one-line Exploratory Data Analysis (EDA) experience in a consistent and fast solution.
- sweetviz. Sweetviz is an open-source Python library that generates beautiful, high-density visualizations to kickstart EDA (Exploratory Data Analysis) with just two lines of code. Output is a fully self-contained HTML application.
- lux APLI. Lux is a Python library that facilitate fast and easy data exploration by automating the visualization and data analysis process. By simply printing out a dataframe in a Jupyter notebook, Lux recommends a set of visualizations highlighting interesting trends and patterns in the dataset.
- DataPrep. DataPrep.EDA is the fastest and the easiest EDA (Exploratory Data Analysis) tool in Python. It allows you to understand a Pandas/Dask DataFrame with a few lines of code in seconds.
- AutoViz. AutoViz performs automatic visualization of any dataset with one line of code. Give it any input file (CSV, txt or json format) of any size and AutoViz will visualize it.

ydata-profiling example.

```
#!pip install visions==0.7.4 --quiet
```

```
!pip install openai --quiet
```

```
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     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 75.9/75.9 kB 5.7 MB/s eta 0:00:0
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     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 58.3/58.3 kB 5.4 MB/s eta 0:00:0
```

```
!pip install tiktoken --quiet
```

```
!pip install cohere --quiet
```

```
!pip install fastapi --quiet
```

```
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 92.1/92.1 kB 2.3 MB/s eta 0:00:0
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 71.5/71.5 kB 4.9 MB/s eta 0:00:0
```

```
!pip install python-multipart --quiet
```

```
!pip install uvicorn --quiet
```

```
!pip install kaleido --quiet
```

```
# pandas-profiling requirements

import sys
!{sys.executable} -m pip install -U ydata-profiling[notebook] --quiet
!jupyter nbextension enable --py widgetsnbextension
```

```
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 357.8/357.8 kB 3.9 MB/s eta 0:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 102.7/102.7 kB 4.5 MB/s eta 0:00
  Preparing metadata (setup.py) ... done
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 686.1/686.1 kB 7.3 MB/s eta 0:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 293.3/293.3 kB 7.8 MB/s eta 0:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 296.5/296.5 kB 7.6 MB/s eta 0:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 4.7/4.7 MB 18.3 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.6/1.6 MB 29.9 MB/s eta 0:00:00
  Building wheel for htmlmin (setup.py) ... done
Enabling notebook extension jupyter-js-widgets/extension...
Paths used for configuration of notebook:
        /root/.jupyter/nbconfig/notebook.json
Paths used for configuration of notebook:

      - Validating: OK
Paths used for configuration of notebook:
        /root/.jupyter/nbconfig/notebook.json
```

```
import numpy as np
import pandas as pd
from ydata_profiling import ProfileReport
```

```
# Read Penguins dataset

pdataset = "https://raw.githubusercontent.com/clizarraga-UAD7/Datasets/main/pengu
df = pd.read_csv(pdataset)
```

```
# General dataframe info
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 344 entries, 0 to 343
Data columns (total 7 columns):
 #   Column             Non-Null Count  Dtype  
---  ------             --------------  -----  
 0   species            344 non-null    object 
 1   island             344 non-null    object 
 2   culmen_length_mm   342 non-null    float64
 3   culmen_depth_mm    342 non-null    float64
 4   flipper_length_mm  342 non-null    float64
 5   body_mass_g        342 non-null    float64
 6   sex                334 non-null    object 
dtypes: float64(4), object(3)
memory usage: 18.9+ KB
```

```
# Drop rows with missing numbers
df=df.dropna()
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 334 entries, 0 to 343
Data columns (total 7 columns):
 #   Column             Non-Null Count  Dtype  
---  ------             --------------  -----  
 0   species            334 non-null    object 
 1   island             334 non-null    object 
 2   culmen_length_mm   334 non-null    float64
 3   culmen_depth_mm    334 non-null    float64
 4   flipper_length_mm  334 non-null    float64
 5   body_mass_g        334 non-null    float64
 6   sex                334 non-null    object 
dtypes: float64(4), object(3)
memory usage: 20.9+ KB
```

```
# Pandas df.describe() summary
df.describe()
```

	culmen_length_mm	culmen_depth_mm	flipper_length_mm	body_mass_g
count	334.000000	334.000000	334.000000	334.000000
mean	43.994311	17.160479	201.014970	4209.056886
std	5.460521	1.967909	14.022175	804.836129
min	32.100000	13.100000	172.000000	2700.000000
25%	39.500000	15.600000	190.000000	3550.000000
50%	44.500000	17.300000	197.000000	4050.000000
75%	48.575000	18.700000	213.000000	4793.750000
max	59.600000	21.500000	231.000000	6300.000000

```
# To generate the standard profiling report

profile = ProfileReport(df, title="Profiling Report")


# displaying the report as a set of widgets

profile.to_widgets()
```

```
/usr/local/lib/python3.10/dist-packages/ydata_profiling/profile_report.py:52
  warnings.warn(
```

Summarize dataset: 32/32 [00:07<00:00, 3.79it/s, 100% Completed]

Generate report structure: 1/1 [00:04<00:00, 4.71s/it] 100%

Overview | Variables | Interactions | Correlations | Missing values | Sample

Variables | species

species

island

island	Distinct	3	Biscoe	1…
Categorical			Dream	1…
HIGH CORRELATION	Distinct (%)	0.9%	Torgersen	4.
	Missing	0		
	Missing (%)	0.0%		
	Memory size	5.2 KiB		

More details

Overview		Categories		Words		Characters	
Max length	9	Total characters	2022	Unique	0	1st row	Torgersen
Median length	6	Distinct characters	13	Unique (%)	0.0%	2nd row	Torgersen
Mean length	6.0538922	Distinct categories	2			3rd row	Torgersen
Min length	5	Distinct scripts	1			4th row	Torgersen
		Distinct blocks	1			5th row	Torgersen
		The Unicode Standard assigns character properties					

```
# The HTML report can be directly embedded in a cell

profile.to_notebook_iframe()
```

Render HTML: 100% 1/1 [00:01<00:00, 1.40s/it]

Overview

Dataset statistics

Number of variables	7
Number of observations	334
Missing cells	0
Missing cells (%)	0.0%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	20.9 KiB
Average record size in memory	64.0 B

Variable types

Categorical	3
Numeric	4

Alerts

`body_mass_g` is highly overall correlated with `culmen_length_mm` and 2 other fields (culmen_length_mm, flipper_length_mm, species)	**High correlation**
`culmen_depth_mm` is highly overall correlated with `flipper_length_mm` and 1 other fields (flipper_length_mm, species)	**High correlation**

Sweetviz example.

```
# Sweetviz install
!pip install sweetviz --quiet
```

```
15.1/15.1 MB 27.2 MB/s eta 0:00:
```

```
import sweetviz as sv

my_dataframe = df.copy()

my_report = sv.analyze(df)
my_report.show_html() # Default arguments will generate to "SWEETVIZ_REPORT.html"
```

```
Done! Use 'show' commands to display/save.  [100%] 00:00 -> (00:00 left)
Report SWEETVIZ_REPORT.html was generated! NOTEBOOK/COLAB USERS: the web brows
```

Note: Google Colab does not open HTML. Please download the HTML to your local computer and open it wiith a browser.

```
# To open results in notebook.
my_report.show_notebook()
```

2.3.1

Get updates, docs & report issues here

Created & maintained by Francois Bertrand
Graphic design by Jean-Francois Hains

DataFrame

	334
	0
	84.7 kb
	7
	3
ASSOCIATIONS	4
	0

DataFrame

1 species

VALUES:	334 (100%)
MISSING:	---
DISTINCT:	3 (<1%)

2 island

VALUES:	334 (100%)
MISSING:	---
DISTINCT:	3 (<1%)

3 culmen_length_mm

VALUES:	334 (100%)	MAX	59.6	RANGE	27.
MISSING:	---	95%	52.0	IQR	9.0
DISTINCT:	163 (49%)	Q3	48.6	STD	5.4
		MEDIAN	44.5	VAR	29.
ZEROES:	---	AVG	44.0		
		Q1	39.5	KURT.	-0.87
		5%	35.7	SKEW	0.04
		MIN	32.1	SUM	14,69

4 culmen_depth_mm

VALUES:	334 (100%)	MAX	21.50	RANGE	8.4
MISSING:	---	95%	20.00	IQR	3.1

Lux API example.

```
# Lux installation

!pip install lux-api --quiet
```

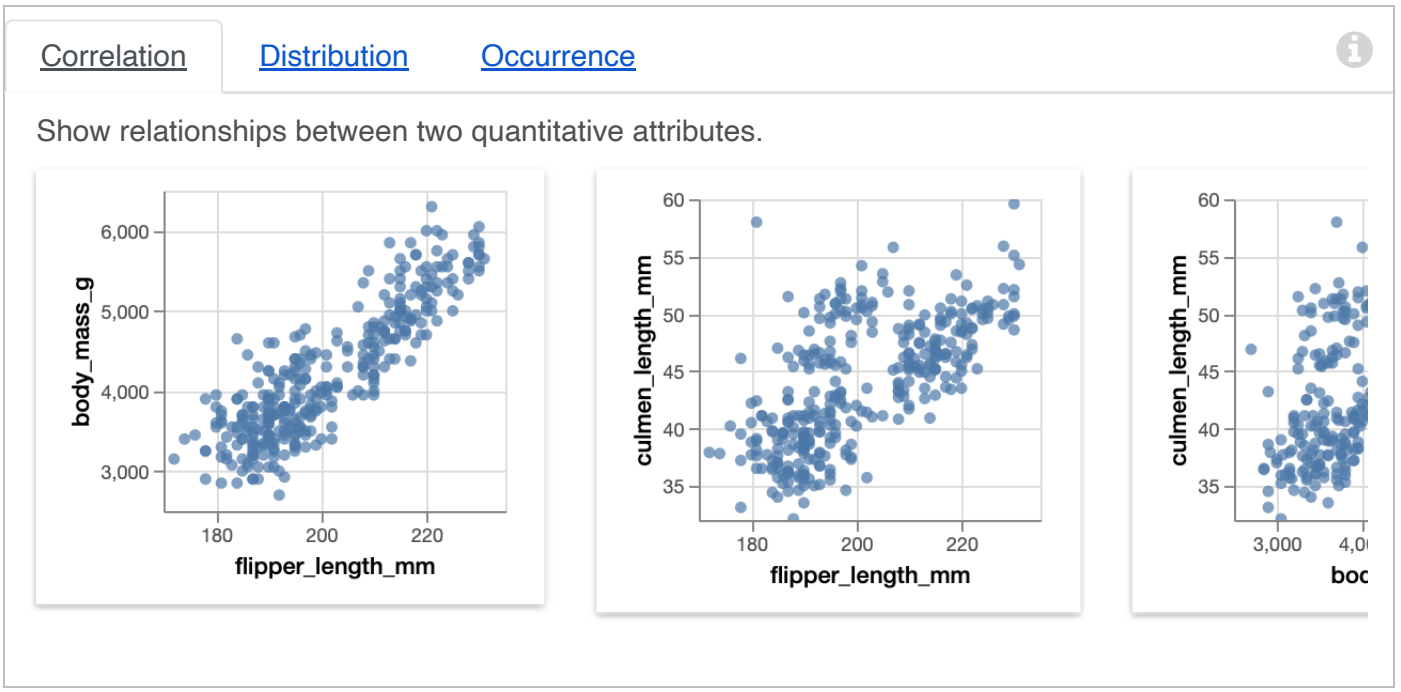
```
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 127.0/127.0 kB 2.8 MB/s eta 0:0
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.5/2.5 MB 34.2 MB/s eta 0:00:0
  Preparing metadata (setup.py) ... done
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 45.3/45.3 kB 4.0 MB/s eta 0:00:0
  Building wheel for lux-api (pyproject.toml) ... done
  Building wheel for lux-widget (setup.py) ... done
```

```
# Activate Jupyter notebook extension

!jupyter nbextension install --py luxwidget
!jupyter nbextension enable --py luxwidget
```

```
Installing /usr/local/lib/python3.10/dist-packages/luxwidget/nbextension/stat
Making directory: /usr/local/share/jupyter/nbextensions/luxwidget/
Copying: /usr/local/lib/python3.10/dist-packages/luxwidget/nbextension/static/
Copying: /usr/local/lib/python3.10/dist-packages/luxwidget/nbextension/static/
Copying: /usr/local/lib/python3.10/dist-packages/luxwidget/nbextension/static/
- Validating: OK

    To initialize this nbextension in the browser every time the notebook (or

         jupyter nbextension enable luxwidget --py

Enabling notebook extension luxwidget/extension...
Paths used for configuration of notebook:
       /root/.jupyter/nbconfig/notebook.json
Paths used for configuration of notebook:

      - Validating: OK
Paths used for configuration of notebook:
       /root/.jupyter/nbconfig/notebook.json
```

```
# Load Lux and add local commands for Googe Colab

import lux
import pandas as pd

from google.colab import output
output.enable_custom_widget_manager()



# Read dataset

pdataset = "https://raw.githubusercontent.com/clizarraga-UAD7/Datasets/main/pengu
df = pd.read_csv(pdataset)

# Drop rows with missing numbers
df=df.dropna()



# Now print df and a Toggle widget will appear to switch between Pandas and Lux

df
```

Toggle Pandas/Lux



DataPrep example

```
!pip install -U git+https://github.com/sfu-db/dataprep.git@develop --quiet
```

```
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 18.5/18.5 MB 37.4 MB/s eta 0:00:
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 133.6/133.6 kB 14.3 MB/s eta 0:0
  Preparing metadata (setup.py) ... done
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.1/3.1 MB 41.4 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.0/1.0 MB 43.1 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.0/1.0 MB 42.1 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.0/3.0 MB 62.0 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 764.0/764.0 kB 16.9 MB/s eta 0:0
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 6.4/6.4 MB 51.8 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 49.6/49.6 kB 4.9 MB/s eta 0:00:0
  Building wheel for dataprep (pyproject.toml) ... done
  Building wheel for sqlalchemy (setup.py) ... done
  Building wheel for metaphone (setup.py) ... done
ERROR: pip's dependency resolver does not currently take into account all the
bigframes 0.20.1 requires sqlalchemy<3.0dev,>=1.4, but you have sqlalchemy 1.3
ipython-sql 0.5.0 requires sqlalchemy>=2.0, but you have sqlalchemy 1.3.24 whi
panel 1.3.8 requires bokeh<3.4.0,>=3.2.0, but you have bokeh 2.4.3 which is in
tiktoken 0.6.0 requires regex>=2022.1.18, but you have regex 2021.11.10 which
ydata-profiling 4.6.4 requires pydantic>=2, but you have pydantic 1.10.14 whic
```

```
from dataprep.eda import *
from dataprep.eda import plot, plot_correlation, plot_missing, plot_diff, create_
```

```
---------------------------------------------------------------------------
ModuleNotFoundError                       Traceback (most recent call last)
<ipython-input-31-57b7790d8da8> in <cell line: 1>()
----> 1 from dataprep.eda import *
      2 from dataprep.eda import plot, plot_correlation, plot_missing,
plot_diff, create_report

7 frames
/usr/local/lib/python3.10/dist-packages/pydantic/_migration.py in
wrapper(name)

ModuleNotFoundError: No module named 'pydantic._internal'
```

```
# Read dataset

pdataset = "https://raw.githubusercontent.com/clizarraga-UAD7/Datasets/main/pengu
df = pd.read_csv(pdataset)

# Drop rows with missing numbers
#df=df.dropna()


plot(df)


# Plot missing values

plot_missing(df)


# Missing value overview

plot_missing(df, "sex")


# Correlation overview

plot_correlation(df)
```

```
# Understand how other columns correlated to the given column

plot_correlation(df, "body_mass_g")


# Numerical column overview

plot(df, "body_mass_g")


# Categorical value overview

plot(df, "sex")


# Numerical vs. Numerical column relationship

plot(df, "flipper_length_mm", "body_mass_g")


# Numerical vs. Categorical column relationship

plot(df, "body_mass_g", "sex")


# Create a full report

create_report(df)
```

AutoViz example

```
# Install AutoViz

!pip install autoviz --quiet
```

```
                                        68.3/68.3 kB 2.2 MB/s eta 0:00:0
                                        421.5/421.5 kB 6.2 MB/s eta 0:00
                                        4.3/4.3 MB 13.5 MB/s eta 0:00:00
                                        3.1/3.1 MB 26.4 MB/s eta 0:00:00
                                        17.3/17.3 MB 35.9 MB/s eta 0:00:
                                        1.9/1.9 MB 66.4 MB/s eta 0:00:00
                                        255.9/255.9 MB 4.5 MB/s eta 0:00
                                        87.3/87.3 kB 9.6 MB/s eta 0:00:0
                                        20.8/20.8 MB 54.6 MB/s eta 0:00:
                                        20.8/20.8 MB 56.7 MB/s eta 0:00:
                                        20.8/20.8 MB 61.1 MB/s eta 0:00:
                                        20.1/20.1 MB 18.4 MB/s eta 0:00:
                                        20.1/20.1 MB 50.5 MB/s eta 0:00:
                                        20.4/20.4 MB 65.6 MB/s eta 0:00:
                                        20.4/20.4 MB 44.8 MB/s eta 0:00:
                                        20.1/20.1 MB 13.5 MB/s eta 0:00:
                                        20.0/20.0 MB 31.8 MB/s eta 0:00:
                                        20.0/20.0 MB 38.5 MB/s eta 0:00:
                                        20.0/20.0 MB 49.3 MB/s eta 0:00:
                                        20.0/20.0 MB 15.3 MB/s eta 0:00:
                                        20.0/20.0 MB 44.5 MB/s eta 0:00:
                                        20.0/20.0 MB 47.1 MB/s eta 0:00:
                                        19.9/19.9 MB 15.6 MB/s eta 0:00:
                                        19.9/19.9 MB 48.3 MB/s eta 0:00:
                                        19.9/19.9 MB 48.2 MB/s eta 0:00:
                                        19.9/19.9 MB 16.9 MB/s eta 0:00:
                                        20.8/20.8 MB 41.8 MB/s eta 0:00:
ERROR: pip's dependency resolver does not currently take into account all the
ydata-profiling 4.6.4 requires pydantic>=2, but you have pydantic 1.10.14 whic
```

```
# Load AutoViz libraries

from autoviz.AutoViz_Class import AutoViz_Class
AV = AutoViz_Class()

# AutoViz does not display plots automatically. The next line is needed.
%matplotlib inline
```

```
Imported v0.1.804. After importing autoviz, you must run '%matplotlib inline'
    AV = AutoViz_Class()
    dfte = AV.AutoViz(filename, sep=',', depVar='', dfte=None, header=0, verbo
               chart_format='svg',max_rows_analyzed=150000,max_cols_analyzed=3
```

```
# Generate visualizations
pdataset = "https://raw.githubusercontent.com/clizarraga-UAD7/Datasets/main/pengu

AV.AutoViz(pdataset)
```

```
Shape of your Data Set loaded: (344, 7)
#############################################################################
######################## C L A S S I F Y I N G  V A R I A B L E S  ##########
#############################################################################
Classifying variables in data set...
    Number of Numeric Columns =  4
    Number of Integer-Categorical Columns =  0
    Number of String-Categorical Columns =  3
    Number of Factor-Categorical Columns =  0
    Number of String-Boolean Columns =  0
    Number of Numeric-Boolean Columns =  0
    Number of Discrete String Columns =  0
    Number of NLP String Columns =  0
    Number of Date Time Columns =  0
    Number of ID Columns =  0
    Number of Columns to Delete =  0
    7 Predictors classified...
        No variables removed since no ID or low-information variables found ir
To fix these data quality issues in the dataset, import FixDQ from autoviz...
    All variables classified into correct types.
```

	Data Type	Missing Values%	Unique Values%	Minimum Value	Maximum Value	DQ Issue
species	object	0.000000	0			No issue
island	object	0.000000	0			No issue

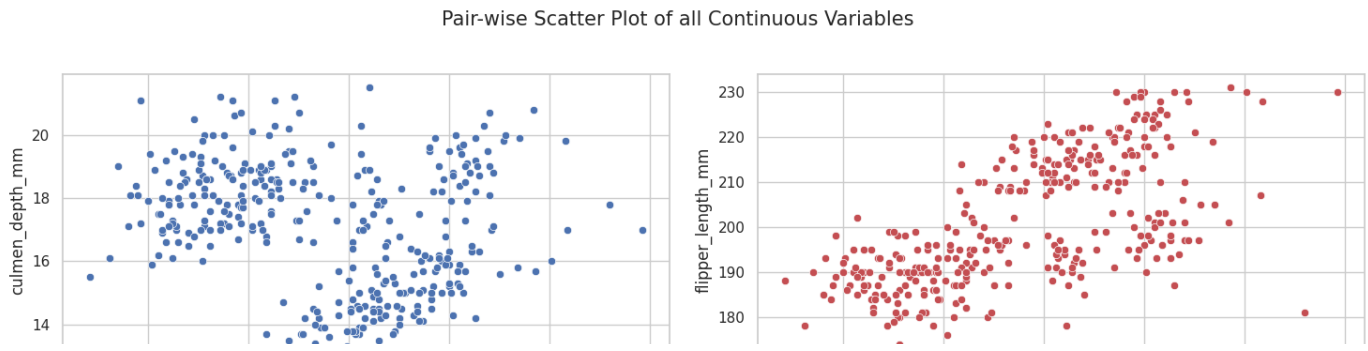
2 missing values.

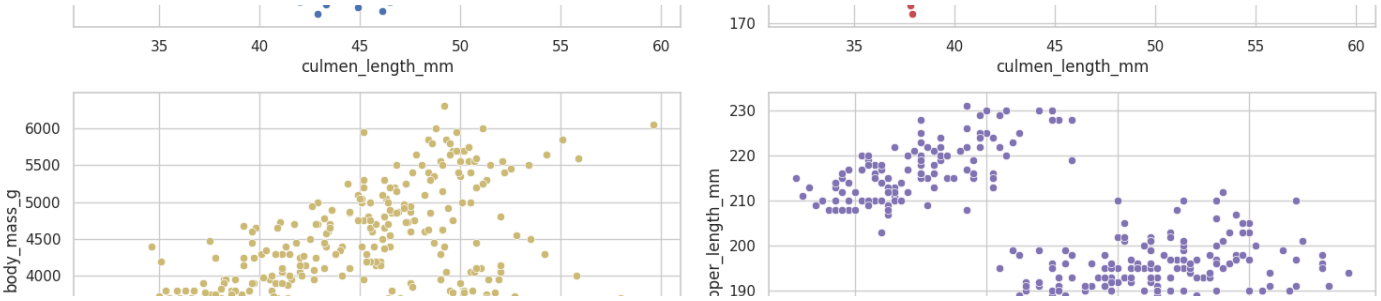
culmen_length_mm	float64	0.581395	NA	32.100000	59.600000	2 missing values. Impute them with mean, median, mode, or a constant value such as 123.
culmen_depth_mm	float64	0.581395	NA	13.100000	21.500000	2 missing values. Impute them with mean, median, mode, or a constant value such as 123.
flipper_length_mm	float64	0.581395	NA	172.000000	231.000000	2 missing values. Impute them with mean, median, mode, or a constant value such as 123.
body_mass_g	float64	0.581395	NA	2700.000000	6300.000000	2 missing values. Impute them with mean, median, mode, or a constant value such as 123., Column has a high correlation with ['flipper_length_mm']. Consider dropping one of them.
						10 missing values. Impute them with mean, median, mode, or a constant value

Start coding or generate with AI.

						them into a single category or drop the categories., Mixed dtypes: has 2 different data types: object, float,

```
Number of All Scatter Plots = 10
```




35
40
45
50
55
60
culmen_length_mm
6000
5500
5000
4500
4000
body_mass_g
170
35
40
45
50
55
60
culmen_length_mm
230
220
210
200
190